Lecture 11

February 12， 2026

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The Market Efficiency Question

- Financial economics asks two related questions:
  - How are expected returns determined?(CAPM, APT, multifactor models)
  - Can investors systematically earn abnormal returns?
- Efficient Market Hypothesis attempts to address the second question by asking:
  - Do competitive markets eliminate predictable profit opportunities?
- Related questions:
  - Do prices adjust quickly to new information?
  - Is there a “free lunch” in financial markets?
  - Can skill reliably outperform competition?

Definition of Market Efficiency

- A market is efficient with respect to an information set I if:

$$P_t = E(P_t | I_t)$$

- Implications:
  - Prices reflect all information in I_t.
  - Price changes occur only when new information arrives.
  - Expected abnormal returns conditional on I_t are zero:

$$E(\alpha_t | I_t) = 0$$

  - Efficiency is always defined relative to an information set.

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Competition and Equilibrium Logic

- Why should markets be efficient?
  - Many rational, profit-seeking investors
  - Rapid information diffusion
  - Low marginal cost of trading
  - Strong incentives to exploit mispricing
- If predictable excess returns exist:
  - investors trade against them
  - prices adjust
  - opportunities disappear
- Efficiency is an equilibrium outcome of competition.

The Fair Game Property

- EMH requires the fair game condition, not a strict random walk.
- Key implications:
  - Prices may be “wrong,” but errors are unpredictable.
  - Returns can be volatile.
  - Risk premia may vary over time.
  - Information arrival drives price changes.
- Fair game ≠ constant prices.
- Fair game ≠ zero volatility.
- Fair game = no predictable excess returns given available information.

- Note that random walk is a stronger condition requiring independence and identical distribution.

Information, Incentives, and the Limits of Efficiency

- Information is the key input to security prices.
  - News about cash flows, discount rates, and risk is constantly arriving.
  - Investors devote substantial resources to discovering and interpreting information.
- Markets can only be efficient because information is actively produced and traded upon.
- Strategies for Acquiring and Processing Information
  - Investors use many tools to uncover value-relevant information:
  - Fundamental analysis of financial statements
  - Macroeconomic forecasting
  - Quantitative models and alternative data
  - Data services (Bloomberg, FactSet, Refinitiv)
  - Expert networks and industry specialists
  - Channel checks and supply-chain analysis
  - Machine learning applied to large unstructured datasets
- The competition to interpret information is central to price discovery.

Information, Incentives, and the Limits of Efficiency

- Regulation and Legal Boundaries
  - Regulation FD (Fair Disclosure):
    - requires material information be disseminated broadly
    - limits selective disclosure to favored investors
  - Material Non-Public Information (MNPI):
    - trading on MNPI is illegal
    - insider trading restrictions limit unfair informational advantage
    - Legal information gathering is encouraged; trading on MNPI is not.
- Suppose all investors were purely passive indexers:
  - no one analyzes firms
  - no one trades on information
  - no one corrects mispricing
  - Would prices still fully reflect information?

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Efficiency Is Not Free (Grossman–Stiglitz)

- If markets were perfectly efficient then $P_t = E(P_t | I_t)$
  - prices would already reflect all available information
  - no investor could earn abnormal returns
  - no incentive would exist to collect costly information
- The Grossman–Stiglitz Paradox
  - Information acquisition is costly.
  - If prices fully reflect all information:
    - informed investors cannot earn excess returns
    - no one would pay to become informed
  - If no one becomes informed prices cannot reflect information.
  - Therefore, perfect informational efficiency is impossible.
  - Markets must be only “efficient enough” to compensate informed traders.

Efficiency Is Not Free (Grossman–Stiglitz)

- Equilibrium Interpretation
  - In equilibrium:
    - Some investors incur costs to gather and process information.
    - They earn expected returns that just compensate those costs.
    - Prices reflect most, but not all, information.
    - Efficiency is therefore an equilibrium outcome of costly information production, not a frictionless ideal.
- Implication
  - Small mispricings may persist.
  - Information production and arbitrage are necessary for efficiency.
  - Passive investing relies on active investors to maintain price accuracy.
  - Markets are competitive —but informational efficiency has limits.

Forms of Market Efficiency

- Market efficiency is always defined relative to an information set I_t.
- Three standard forms:
  - Weak Form: Prices reflect all information in past prices and returns.
  - Semi-Strong Form: Prices reflect all publicly available information.
  - Strong Form: Prices reflect all information, public and private.
- Each form implies progressively stronger restrictions on abnormal return opportunities.

Weak-Form Efficiency

- Weak-form EMH asserts:
  - $E(R_t+1|R_t,R_t-1,\ldots)=0$
  - Past price patterns cannot be used to predict abnormal risk-adjusted returns.
  - Stock prices already reflect all information that can be derived by examining trading data.
- Implications:
  - Technical analysis should not systematically add value.
  - Price changes respond only to new information, not predictable trends.
- Weak-form efficiency is the minimal notion of informational efficiency.

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Empirical Evidence: Serial Correlation

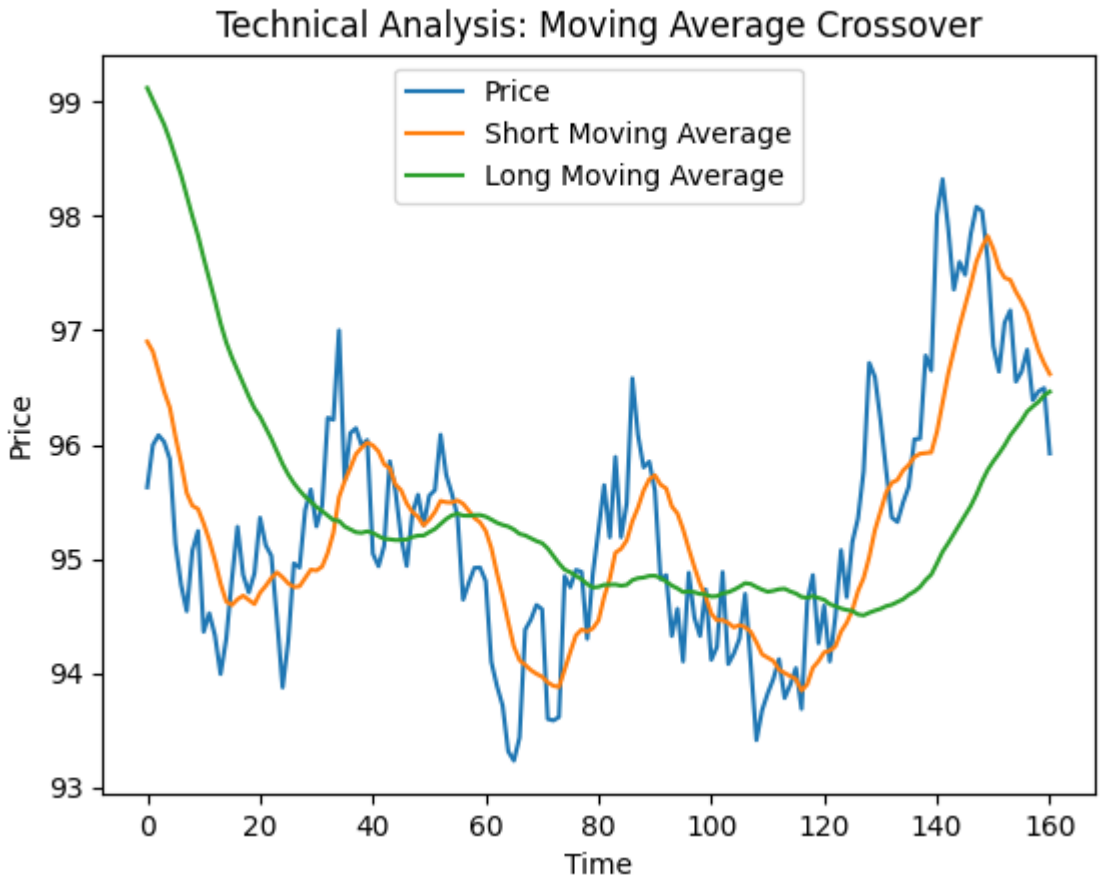
- Tests of weak-form efficiency examine:

$$Corr(\mathrm{R}_t, \mathrm{R}_{t-k}) \text{ for } k>0$$

- Findings:
  - daily and weekly autocorrelations are near zero
  - bid–ask bounce can generate small negative correlations
  - longer-horizon reversals exist but are economically modest
- Conclusion:
  - return predictability is limited
  - trading costs often eliminate exploitable profits
  - Weak-form efficiency is broadly supported.

Technical Analysis and Trading Rules

- Technical analysis attempts to forecast returns using only:
  - past prices
  - past returns
  - trading volume and chart patterns
- Common tools:
  - **Momentum / Relative Strength**: buy recent winners, sell recent losers
  - **Moving Averages**: buy when short-term MA crosses above long-term MA
  - **Relative Strength Index (RSI)**: measures overperformed/underperformed conditions
  - **Market Breadth**: number of advancing vs declining stocks
- Weak-form predicts rules based solely on price history should not generate consistent abnormal profits.



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Evidence on Technical Trading

- Empirical studies of return predictability find:
  - Very small short-horizon autocorrelations.
  - Slight negative correlation at very short horizons (bid–ask bounce).
  - Some evidence of intermediate-term momentum (3–12 months).
  - Long-horizon reversals over multi-year periods.
- However:
  - Individual stock returns remain highly volatile and difficult to predict.
  - Momentum effects are weak at the individual security level.
  - Predictability is more evident in diversified portfolios formed on past returns.
  - After transaction costs and risk adjustment, economic significance is reduced.

Semi-Strong Form Efficiency

- Semi-strong EMH asserts:

$$E(R_{t+1}|\text{all public information})=E(R_{t+1})$$

- Public information includes:
  - earnings announcements
  - financial statements
  - macroeconomic releases
  - analyst research
  - corporate news
- Implication is that fundamental analysis should not reliably generate abnormal risk-adjusted returns.

Event Studies: Measuring Market Response to News

- Semi-strong efficiency is commonly tested using event studies, which examine how quickly prices incorporate new public information.
- An event study focuses on a specific announcement date:
  - earnings releases
  - merger announcements
  - regulatory actions
  - macroeconomic news
- Event studies start with looking at normal returns
- The abnormal return is the portion of the realized return not explained by a benchmark model:

$$AR_t = R_t - E(R_t)$$

- where: R_t: realized return on the security
- $E(R_t)$: expected return absent the even (often estimated using CAPM or a market model)

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Event Studies: Measuring Market Response to News (con.)

- However, concerns around leakage arise, leading to the measure of cumulative abnormal returns.
- This attempts to measure the total price impact over an event window:

$$CAR(\tau_1, \tau_2) = \sum_{t=\tau_1}^{\tau_2} AR_t$$

- CAR captures whether abnormal returns persist:
  - before the announcement (information leakage)
  - at the announcement (immediate adjustment)
  - after the announcement (delayed reaction)
- Semi-strong efficiency predicts:
  - abnormal returns occur immediately when news is released
  - little or no systematic drift afterward

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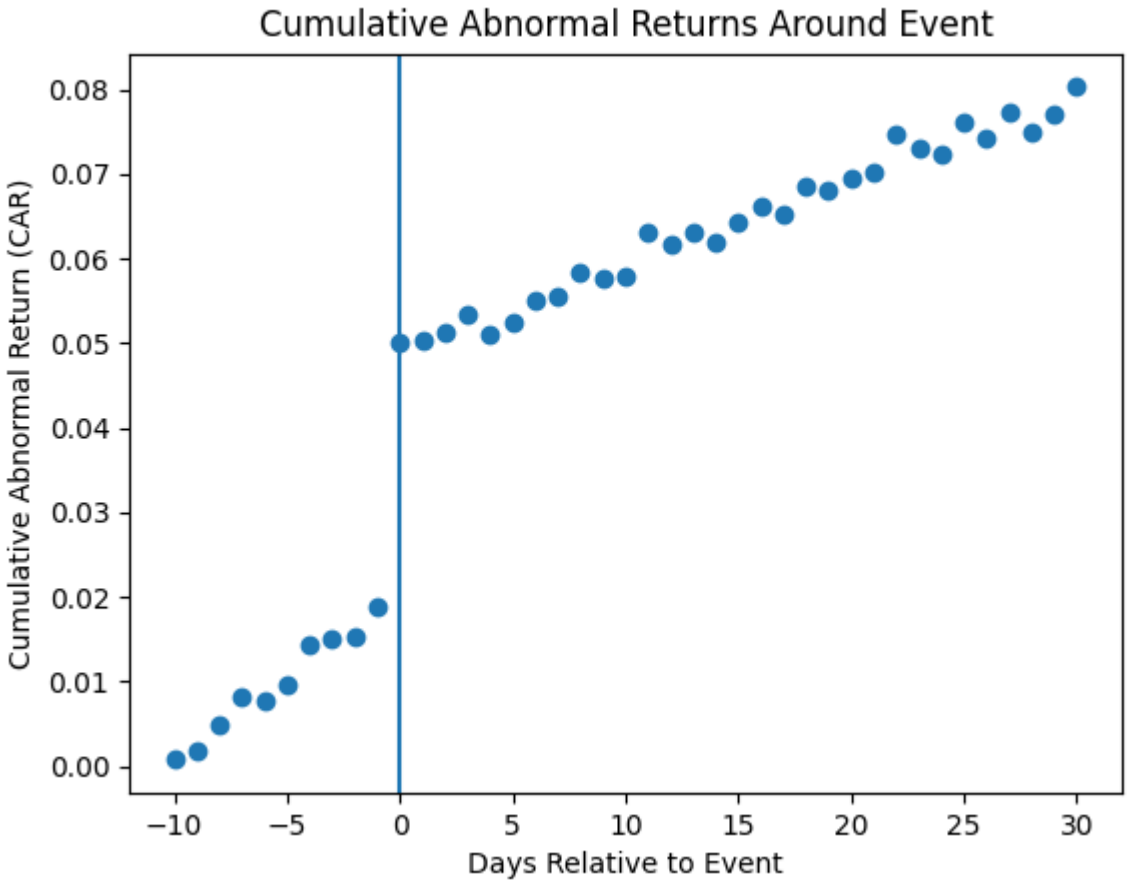
Post-Event Drift and Market Efficiency

- Event studies often examine whether abnormal returns persist after announcements.
- Under semi-strong efficiency:

$$CAR(\tau_1,\tau_2)\approx 0 \text{ for } \tau_1>0$$

- after the announcement window.
- However, empirical evidence shows:
  - Post-Earnings Announcement Drift (PEAD)
  - gradual continuation of abnormal returns following earnings surprises
  - small but statistically significant delayed adjustment



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Strong-Form Efficiency

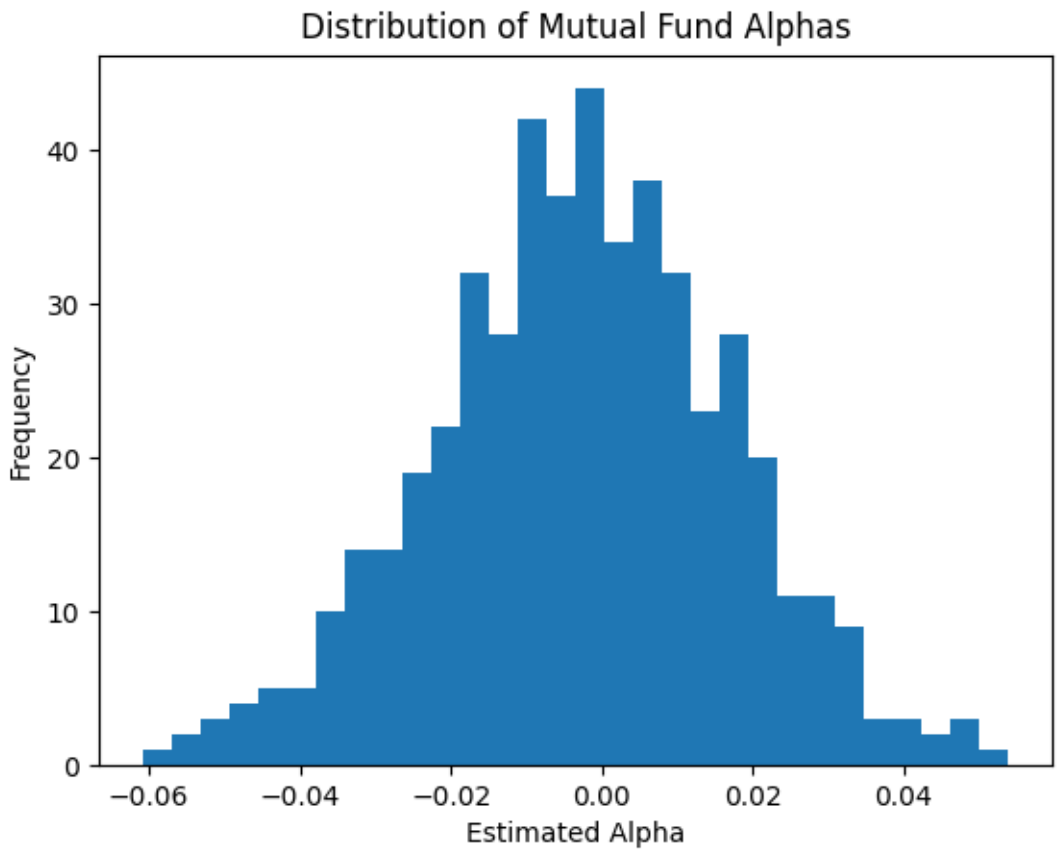
- Strong-form EMH asserts:
  - prices reflect all information, public and private
- Implication:
  - even insiders cannot earn abnormal returns consistently
- Evidence:
  - corporate insiders do earn abnormal returns
  - some institutional investors obtain informational advantages
- Conclusion:
  - strong-form efficiency is not supported in practice
  - informational advantages can persist, especially when information is costly or restricted

Active Management and Market Efficiency

- Semi-strong efficiency implies:
  - publicly available information should already be reflected in prices
  - abnormal risk-adjusted returns should be difficult to achieve consistently
- Active management is a zero-sum game before costs
  - after fees, trading costs, and price impact, it becomes negative-sum
- This means the average active investor must underperform passive benchmarks net of costs
- Persistent alpha should be rare in competitive markets

Mutual Fund Evidence: Alpha and Persistence

- Empirical findings show:
  - estimated alphas cluster near zero
  - limited evidence that past winners remain winners
  - stronger persistence among poor performers (high fees, weak strategies)
- Observed outperformance often reflects factor tilts or luck
- Competition and costs make sustained abnormal performance difficult



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Interpreting Alpha: Magnitude, Bias, and Luck

- Evidence suggests that average mutual fund alpha is close to zero.
- However, several considerations complicate the interpretation.
- Small Alpha Can Still Be Economically Meaningful
  - Even modest abnormal performance can generate large dollar gains at scale:
    - An annual alpha of 0.3% on a $5 billion fund implies: $0.003 \times 5{,}000{,}000{,}000 \approx \$15$ million per year
    - Economically meaningful skill may appear statistically small
    - noise makes small alphas difficult to detect reliably
  - This is the magnitude problem.

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Interpreting Alpha: Magnitude, Bias, and Luck (con.)

- Selection and Reporting Bias
  - The best strategies may not appear in mutual fund databases:
    - successful managers may operate in hedge funds or private vehicles
    - proprietary strategies may remain undisclosed to preserve profitability
  - Observed mutual fund performance may understate the upper tail of skill.
- Luck vs Skill Is Difficult to Separate
  - In any large universe of funds some will outperform persistently by chance alone
  - Even if some managers are truly skilled, distinguishing skill from luck requires long samples
  - Statistical inference is challenging
  - Thus, observed outperformance is not definitive evidence of skill, but lack of clear alpha is not proof that skill is absent

Back To The Joint Hypothesis Problem

- Tests of market efficiency are inherently joint tests of two hypotheses:
  - Market efficiency
  - The asset pricing model used to define “normal” expected returns
- Abnormal Returns Require a Benchmark
  - To measure alpha, we compute: $\alpha_t = R_t - E(R_t|\text{model})$
  - But expected returns are not directly observable.
  - They must be estimated using a model such as:
    - CAPM
    - multifactor models (Fama–French)
    - consumption-based models

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Back To The Joint Hypothesis Problem (con.)

- Interpretation Challenge
  - If we observe nonzero abnormal returns:
    - Are markets inefficient?
    - Or is the benchmark model incomplete?
    - Example:
      - CAPM may generate apparent alpha
      - but a multifactor model may explain it as compensation for omitted risks
    - “Anomaly” may reflect mispricing — or model misspecification.
  - Empirical evidence cannot cleanly distinguish:
    - inefficiency
    - missing risk factors
    - behavioral effects
    - statistical noise
  - This is why debates over efficiency and anomalies remain unresolved.

Market Anomalies

- Empirical patterns that challenge simple efficiency include:
  - Momentum (intermediate horizons)
  - Value premium (HML)
  - Size effect
  - Post-earnings announcement drift
  - Long-horizon reversals
- These patterns have been documented repeatedly across markets.
  - Do they represent inefficiency?
  - Or compensation for omitted risk?
- Anomalies are central to the debate over market efficiency.

Introduction to Behavioral Finance

- Behavioral finance relaxes the assumption of fully rational investors.
- Key questions:
  - Do systematic psychological biases affect decisions?
  - Can these biases generate predictable return patterns?
  - Why might prices deviate from fundamental value?
- Behavioral finance complements EMH:
  - explains why mispricing may arise
  - explains why it may persist
- Some believe that anomalies are consistent with “irrationalities” that seem to characterize individuals making complicated decisions.

Two “Irrationalities”

- These irrationalities fall into two broad categories:
  - Information Processing
  - Behavioral Biases
- But would irrationalities be sufficient to render capital markets inefficient?
  - First, we must assume these irrationalities were sufficient to affect prices
  - If so, could arbitrageurs take advantage?
- Second leg of behavioral finance posits that the actions of such arbitrageurs are limited and inadequate to push prices to intrinsic value
  - If prices are right, general consensus that no easy profit opportunities
  - However, converse is not true
- Important as we think about active vs passive investing
  - Even if active mangers fail to outperform passive strategies, this, in and of itself, does not mean markets are efficient

Errors in Information Processing

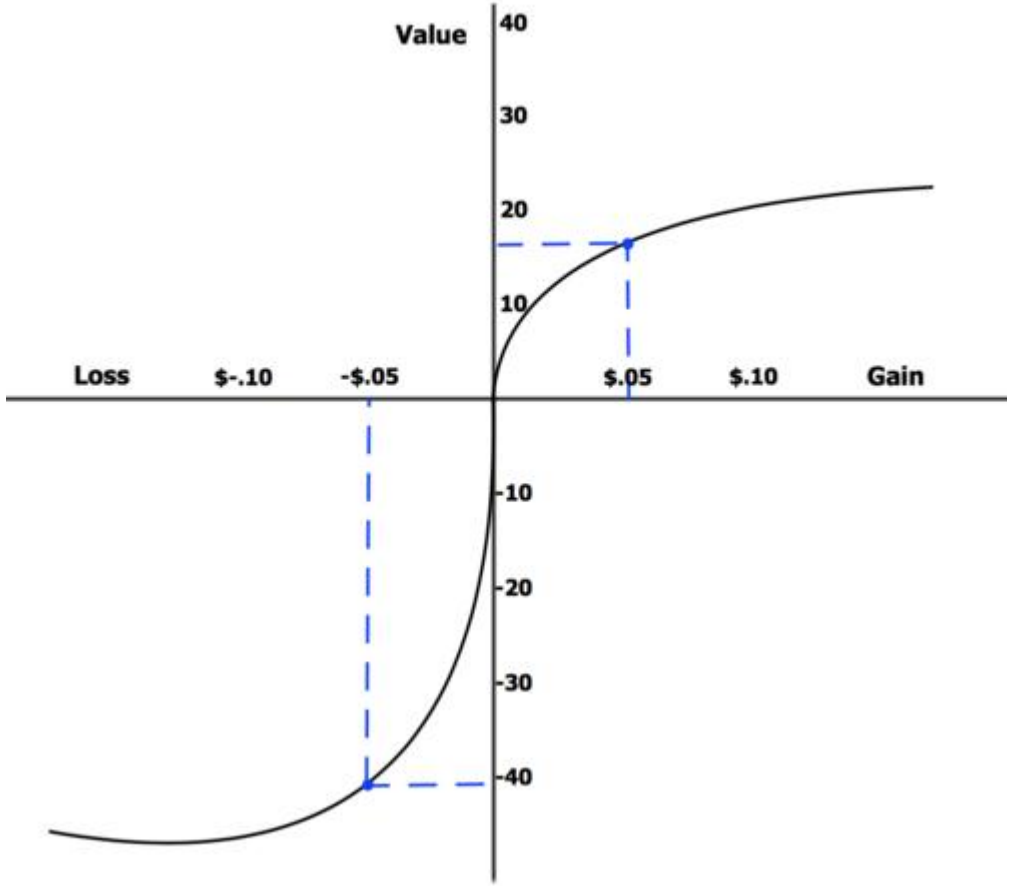
- Investors may process information imperfectly.
- Common errors include:
  - Limited Attention: investors overlook less salient information
  - Underreaction: slow incorporation of new information
  - Overreaction: excessive response to recent news
  - Conservatism Bias: insufficient updating of prior beliefs
  - Confirmation Bias: overweight information that confirms existing beliefs
  - Extrapolation / Pattern Recognition: assume recent trends will continue indefinitely
- These mechanisms can generate momentum and reversals.

Behavioral Biases in Preferences

- Investor behavior may depart from standard expected utility.
- Examples:
  - Framing Effects: decisions depend on presentation of outcomes
  - Mental Accounting: investors treat money in separate “accounts”
  - Regret Avoidance: avoid actions that could lead to future regret
  - Affect / Feelings: emotions influence risk perception
- Preferences are not always stable or rational.

Prospect Theory

- Prospect theory modifies expected utility:
  - outcomes are evaluated relative to a reference point
  - losses loom larger than gains
- Value function:
  - concave for gains
  - convex for losses
  - steeper for losses



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Prospect Theory (con).

- Implications
  - In the domain of gains:
    - when investors face a risky opportunity to earn profits, they typically prefer a smaller but guaranteed gain over a gamble with a higher expected payoff.
    - This reflects diminishing sensitivity to additional gains (concavity).
  - In the domain of losses:
    - when investors face a risky choice involving losses, they often become willing to gamble to avoid locking in a sure loss, even if the gamble has a lower expected value.
    - This reflects risk-seeking behavior when outcomes are negative (convexity).
  - Asymmetric reactions to good vs bad news
  - Disposition effect and reluctance to realize losses

With This, How To Interpret Anomalies

- Observed return patterns can reflect:
  - Risk-based explanations
    - exposure to distress, liquidity, or macro risk
    - compensation for bad-state performance
  - Behavioral explanations
    - underreaction or overreaction
    - extrapolation and sentiment
  - Model misspecification
    - missing risk factors
    - incorrect benchmark
  - Because of the joint hypothesis problem, anomalies do not automatically imply inefficiency.

Why Don't Traders Eliminate Anomalies Immediately?

- Even when investors believe prices deviate from fundamentals, trading against mispricing is constrained by:
  - fundamental risk (the mispricing might not correct soon)
  - model risk (valuation is uncertain)
  - implementation costs (turnover, market impact, financing)
  - short-sale constraints and borrow costs
  - interim-loss risk (forced liquidation before convergence)
- After costs and risk adjustment economic profitability may be modest
- An anomaly is not the same as a free arbitrage opportunity.
  - they require bearing risk
  - they require capital
  - they may involve shorting and leverage
  - payoffs are uncertain and can be negative for long periods

Examples of Limits to Arbitrage

- Equity Carve-Outs
  - Parent firm sometimes valued less than stake in subsidiary.
  - Apparent negative stub value.
  - Hard to arbitrage due to short-sale and execution constraints.
- Closed-End Fund Discount
  - Fund shares trade at discount to NAV.
  - Rational traders cannot force convergence easily.
  - Discounts persist for long periods.
- These examples illustrate real-world limits to arbitrage.

Investor Sentiment

- Sentiment reflects broad market optimism or pessimism.
- Proxies include:
  - IPO volume
  - First-day IPO returns
  - Closed-end fund discounts
  - Mutual fund flows
  - VIX and option-implied volatility
- This is meaningful in not just trading, but new issues!
  - Insider coverage of deals
  - Cross-over deals

Market Sentiment Indicators

- Investor sentiment is difficult to observe directly, but several market-based measures proxy for optimism or fear.
- Common sentiment indicators include:
  - TRIN Statistic (Arms Index)
    - The TRIN combines price movement and trading volume:

$$TRIN = \frac{\text{Advancing Issues}/\text{Declining Issues}}{\text{Advancing Volume}/\text{Declining Volume}}$$

    - Interpretation:
      - $TRIN$>1: selling pressure dominates (bearish sentiment)
      - $TRIN$<1: buying pressure dominates (bullish sentiment)

Market Sentiment Indicators (con.)

- Barron's Confidence Index
  - Measures perceived credit risk:

$$\text{Confidence Index} = \frac{\text{Yield on High-Grade Bonds}}{\text{Yield on Intermediate-Grade Bonds}}$$

  - Interpretation:
    - Index near 1 → investors confident, low credit risk premium
    - Index falling → rising risk aversion and economic concern
- Put–Call Ratio
  - Measures demand for downside protection:

$$\text{Put–Call Ratio} = \frac{\text{Put Option Volume}}{\text{Call Option Volume}}$$

  - Interpretation:
    - High ratio → pessimism, hedging demand, fear
    - Low ratio → optimism, speculative sentiment

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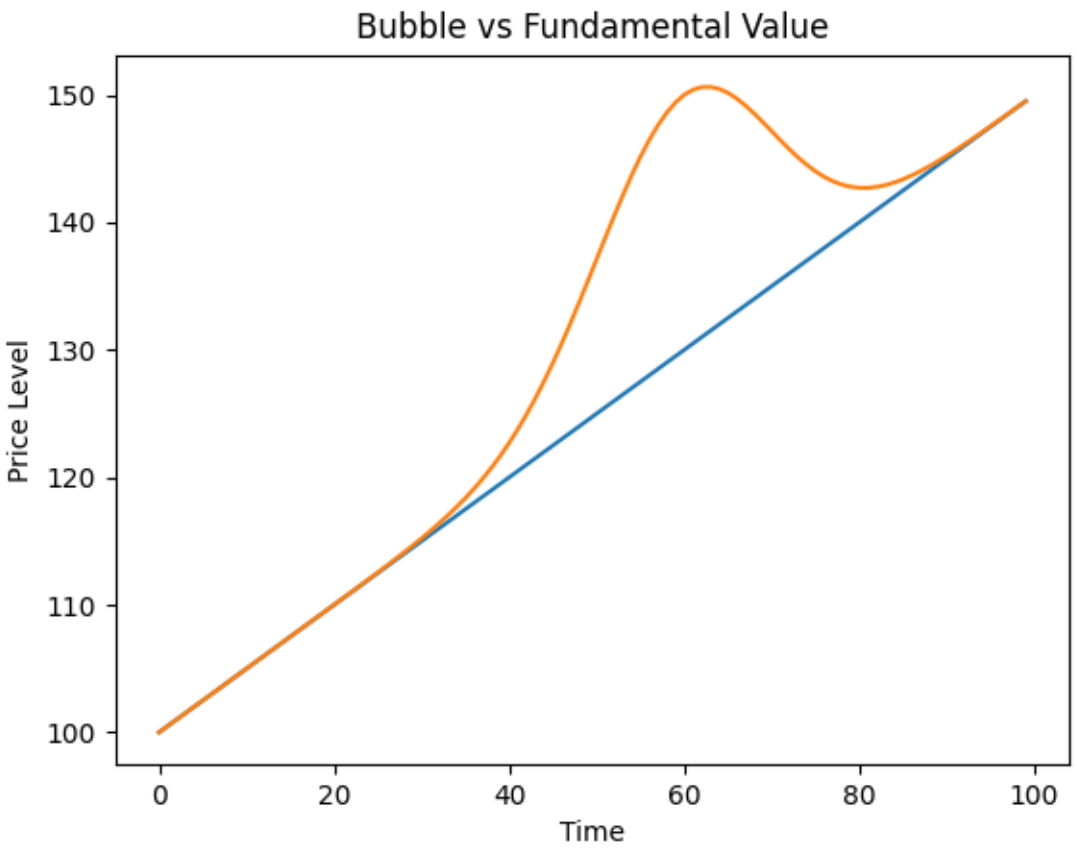
Sentiment and Return Predictability

- If sentiment pushes prices away from fundamentals, we should observe predictable reversals.
- Empirical findings suggest:
  - Periods of high investor sentiment are often followed by lower subsequent average returns, especially among speculative, hard-to-arbitrage stocks.
  - Periods of low sentiment are often followed by higher subsequent returns.
- Possible mechanisms:
  - Overoptimism leads to temporary overpricing.
  - Short-sale constraints prevent immediate correction.
  - Prices gradually revert toward fundamentals.
- Sentiment is time-varying and interacts with limits to arbitrage.

Bubble Dynamics: Prices vs Fundamentals

- A bubble occurs when:
  - market prices rise substantially above fundamental value
  - price increases are driven by expectations of resale rather than cash flows (so capital gains)
  - The bubble component is sustained by:
    - investor optimism
    - belief in continued price appreciation
    - positive feedback trading



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Positive Feedback and Speculative Trading

- Positive feedback traders:
  - buy when prices rise
  - sell when prices fall
- This creates:
  - self-reinforcing price movements
  - trend amplification
  - increased volatility
- As prices rise:
  - more investors enter
  - narratives strengthen
  - valuation discipline weakens
- Momentum can evolve into speculative excess.
- This occurs outside of bubbles – think retail investors in thinly traded stocks

Identifying Bubbles And Persistence

- Bubbles are difficult to diagnose because:
  - fundamental value is uncertain
  - technological or macro shifts may justify high prices
  - ex post collapse does not prove ex ante irrationality
- Even if some investors suspect overvaluation:
  - short-sale constraints limit correction
  - arbitrage is risky and capital constrained
  - prices may rise further before collapsing (think fundamental risk here)
  - rational traders face interim-loss risk
- Thus:
  - mispricing can expand before it contracts
  - bubbles can persist longer than expected
- Limits to arbitrage interact with sentiment.

Are Markets Efficient? A Balanced View

- The evidence suggests:
  - markets respond rapidly to public information
  - consistent abnormal profits are difficult after costs
  - most active managers underperform net of fees
- This leads one to believe EMH is a powerful benchmark for thinking about prices
- However:
  - anomalies exist
  - sentiment can affect prices
  - limits to arbitrage prevent perfect correction
- Market efficiency is best viewed as an approximation, not an absolute.

Efficiency as an Equilibrium Outcome

- Efficiency is not automatic.
- Prices reflect information because:
  - investors compete aggressively
  - information is costly to acquire
  - trading pushes prices toward fair value
- But perfect efficiency is impossible:
  - Grossman–Stiglitz: information must be rewarded
  - arbitrage is risky and constrained
- Markets are “efficient enough” that easy profit opportunities are rare.

The Adaptive Markets Perspective

- Market efficiency may vary over time.
- Adaptive Markets Hypothesis:
  - competition drives efficiency
  - but investor behavior evolves
  - market structure changes
  - strategies decay as they become crowded
- Anomalies may appear, weaken, or disappear; however, efficiency is not fixed, but dynamic.

Implications for Investors

- Key implications of EMH and behavioral finance:
  - Passive investing is difficult to beat net of costs
  - Diversification remains essential
  - Apparent alpha may reflect factor exposure
  - Behavioral biases can lead to poor individual decisions
  - Sentiment-driven mispricing may persist but is hard to exploit
- Markets are hard to outperform consistently, not impossible.

Active vs Passive Management Revisited

- Active management may be justified when:
  - investors have informational advantages
  - markets are illiquid or segmented
  - constraints limit arbitrage
  - specialized skill exists
- But for most investors:
  - broad indexing is a robust baseline
  - fees and turnover are major drags
  - performance persistence is weak

What Does EMH Really Imply?

- EMH does not imply:
  - prices are always correct
  - bubbles cannot occur
  - no one ever earns alpha
- EMH does imply:
  - abnormal profits are difficult to achieve systematically
  - mispricing is hard to identify in real time
  - competition quickly incorporates widely available information
- Efficiency is a benchmark, not a claim of perfection.